FED. SUP CLASS 2620

THE TIRE SHALL BE IN ACCORDANCE WITH THE APPLICABLE REQUIREMENTS OF MIL-T-5041 EXCEPT AS SPECIFIED HEREIN.

SIZE	PLY RATING	STATIC LOAD RATING LBS.	INFL. PRESS. PSI RATED	BURST PRESS. PSI MIN.	BEAD WIDTH IN. MAX.	WEIGHT LBS. MAX.	STATIC UNBAL +1/2 OZ.	TREAD	MOLD SKID DEPTH MIN.	DEFLC +3% -4%
36x11	1/28TL	31,500	290	2/1160	2.80	99	30	3/RIB	.30	32

1/ TL TUBELESS (B)
2/ NEW TIRE
3/ AT LEAST THREE BUT NOT MORE THAN SEVEN RIBS. THE TREAD GROOVES SHALL BE SHAPED INSOFAR AS PRACTICABLE TO PREVENT FOREIGN OBJECTS FROM BEING TRAPPED BETWEEN THE RIBS.

THE TIRE COVERED BY THIS DRAWING SHALL BE SUITABLE FOR USE AND PROVIDE REASONABLE SERVICE LIFE DURING ALL NORMAL OPERATIONS AT TAKEOFF AND LANDING SPEEDS INDICATED HEREIN ON ALL TYPES OF RUNWAYS AND ON AIRCRAFT CARRIERS.

TEST INFLATION PRESSURES:

(1) NEW TIRE DEFLECTION (D) IN INCHES SHALL BE DETERMINED ON A FLAT PLATE AT 31,000 POUND LOAD AND 290 PSI INFLATION PRESSURE. DEFLECTION (D) VALUE SHALL BE USED IN DETERMINING (2) AND (3) BELOW.

(2) INFLATION PRESSURE FOR TESTS A AND B. THAT PRESSURE REQUIRED TO SUPPORT A 31,500 POUND LOAD AT DEFLECTION (D) ON A TEST ARC EQUIVALENT TO FLYWHEEL ARC.

(3) INFLATION PRESSURE FOR TEST C AND D. THAT PRESSURE REQUIRED TO SUPPORT A 34,000 POUND LOAD AT DEFLECTION (D) ON A TEST ARC EQUIVALENT TO FLYWHEEL ARC.

THE QUALIFICATION TIRE SHALL BE TESTED AS FOLLOWS:

TIRE 1: THE TEST TIRE SHALL WITHSTAND 30 CYCLES OF TEST A, 50 CYCLES OF TEST B, 20 CYCLES OF TEST C, 1 CYCLE OF TEST D, THEN BE SUBJECTED TO THE HYDROSTATIC BURST TEST (TEST E).

TEST A - NORMAL TAKEOFF - THE TIRE SHALL BE TAXIED ON THE FLYWHEEL AT 40 MPH FOR 10,000 FEET WITH 31,500 POUNDS LOAD. UPON COMPLETION OF THE TAXI ROLL, THE FLYWHEEL SHALL BE STOPPED AND THEN IMMEDIATELY ACCELERATED AT AN AVERAGE RATE OF 5.9 FT/SEC/SEC FROM 0 TO 200 MPH. THE TIRE SHALL BE UNLANDED AFTER A TAKEOFF ROLL DISTANCE OF 7 300 FEET HAS BEEN COVERED IN APPROXIMATELY 49-50 SECONDS. THE INITIAL TAKEOFF LOAD OF 31,500 POUNDS SHALL BE DECREASED LINEARLY WITH TIME TO 28,000 POUNDS AT 20 SECONDS AFTER THE START OF THE TAKEOFF ROLL, THEN LINEARLY DECREASED TO ZERO LOAD AT THE END OF THE TAKEOFF ROLL.

TEST B - LANDING - THE TIRE SHALL BE LANDED AGAINST THE FLYWHEEL ROTATING AT A PERIPHERAL SPEED OF 160 MPH. THE AVERAGE DECELERATION RATE SHALL BE 5.4 FEET/SEC/SEC FROM 160 MPH TO 40 MPH. THE LANDING ROLL DISTANCE TO 40 MPH SHALL BE 4,700 FEET AND SHALL BE COVERED IN 32-33 SECONDS. THE INITIAL TIRE LOAD SHALL BE LINEARLY INCREASED TO 22,000 POUNDS WITHIN 3 TO 4 SECONDS AFTER TOUCHDOWN. AFTER THE LANDING ROLL OF 4,700 FEET, THE TIRE SHALL CONTINUE TO ROLL FROM THAT POINT AT 40 MPH AND 22,000 POUNDS UNTIL AN ADDITIONAL 10,000 FEET HAVE BEEN COVERED. AT THIS TIME THE TIRE SHALL BE UNLANDED.

TEST C - OVERLOAD TAKEOFF - THE TIRE SHALL BE TAXIED ON THE FLYWHEEL AT 40 MPH FOR 10,000 FEET WITH 34,000 POUNDS LOAD. UPON COMPLETION OF THE TAXI ROLL, THE FLYWHEEL SHALL BE STOPPED AND THEN IMMEDIATELY ACCELERATED AT AN AVERAGE RATE OF 5.9 FT/SEC/SEC FROM 0 TO 200 MPH. THE TIRE SHALL BE UNLANDED AFTER A TAKEOFF ROLL DISTANCE OF 7,300 FEET HAS BEEN COVERED IN 49 TO 50 SECONDS. THE LOAD SHALL BE DECREASED LINEARLY WITH TIME FROM 34,000 LBS TO 30,000 LBS DURING THE FIRST 20 SECONDS OF THE TAKEOFF ROLL AND THEN FURTHER REDUCED LINEARLY WITH TIME TO ZERO LOAD AT THE END OF THE TAKEOFF ROLL.

TEST D - CAMBER TESTS - ONE CYCLE OF TEST D SHALL CONSIST OF 25 CYCLES OF TEST D_1 AND 25 CYCLES OF TEST D_2.

D_1 - OUTBOARD CAMBER (25 CYCLES)

1. ACCELERATE FLYWHEEL TO 20 MPH AND MAINTAIN THIS SPEED THROUGH THE TEST.
2. LAND THE TIRE AGAINST THE FLYWHEEL IN A POSITION OF 15° CAMBER AND LOAD AS FOLLOWS:
   A. INCREASE THE CARRIAGE LOAD TO 25,000 LBS DURING THE FIRST 2 SECONDS AND THEN TO 41,000 LBS WITHIN 3.5 TO 4 SECONDS AFTER LANDING.
   B. MAINTAIN THE 41,000 LBS LOAD AT 20 MPH UNTIL A DISTANCE OF 2,400 FEET HAS BEEN COVERED.

(A) ENTIRE STANDARD REVISED (B) DENOTES CHANGES

APPROVED 15 SEP 1964 REVISED (A) 31 JAN 78 (B) 14 MAY 79

This military standard is approved by NAVAL AIR SYSTEMS COMMAND, Department of the Navy and shall be used by their activity. All other military activities are required to employ this standard where suitable.

P.A. NAVY - AS Other Cust	TITLE: TIRE, PNEUMATIC, AIRCRAFT 36x11 HIGH SPEED, TYPE VII (NAVY)	MILITARY STANDARD MS90346(AS)
PROCUREMENT SPECIFICATION	SUPERSEDES:	SHEET 1 OF 2

DD FORM 1 MAR 72 672-1 (Limited coordination) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE. PROJECT NO. 2620-N188 PLATE NO 23071

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D_2 - INBOARD CAMBER (25 CYCLES)

SAME AS TEST D_1 EXCEPT THAT THE LOAD SHALL BE APPLIED FROM A DIRECTION TO PROVIDE FOR 15° INBOARD CAMBER.

TEST E - BURST TEST - THE TIRE SHALL BE SUBJECTED TO A HYDROSTATIC BURST TEST. THE PRESSURE SHALL BE INCREASED UNTIL THE TIRE FAILS. THE FAILING PRESSURE DESCRIPTION OF FAILURE AND LOCATION SHALL BE REPORTED IN THE QUALIFICATION TEST REPORT.

AIR RETENTION - THE TIRE SHALL BE INFLATED TO A PRESSURE OF 450 PSI FOR AIR RETENTION TESTING IN ACCORDANCE WITH THE REQUIREMENTS AND TEST METHODS OF SPECIFICATION MIL-T-5041

DIMENSIONS - A NEW TIRE SHALL BE MEASURED IN ACCORDANCE WITH THE REQUIREMENTS AND METHODS OF MIL-T-5041 AT THE RATED INFLATION PRESSURE, 290 PSI. IN ADDITION TO THE DIMENSIONS REQUIRED AT RATED INFLATION PRESSURE THE DIMENSIONS OF THE TIRE INFLATED TO 450 PSI (STRETCHED FOR A MINIMUM OF 12 HOURS) SHALL BE DETERMINED AND REPORTED.

INFLATED PROFILE - A SKETCH OF THE TIRE PROFILE AT THE RATED INFLATION PRESSURE AND 450 PSI SHALL BE PREPARED AND INCLUDED AS PART OF THE QUALIFICATION TEST REPORT.

MARKING - IN ADDITION TO THE MARKING REQUIRED BY SPECIFICATION MIL-T-5041, THE DRAWING NUMBER MS 90346 SHALL BE ENGRAVED OR EMBOSSED ON ONE SIDEWALL IN THE VICINITY OF THE SIZE AND PLY RATING MARKINGS.

QUALIFICATION TEST REPORT - THE QUALIFICATION TEST REPORT SHALL LIST THE RESULTS OF ALL QUALIFICATION TESTS AND CONSTRUCTION DETAILS OF THE QUALIFICATION TEST SAMPLE IN THE GENERAL FORM SHOWN IN FIGURE 6 OF MIL-T-5041 WITH DIMENSIONS LISTED AT RATED INFLATION AND 450 PSI. THE REPORT SHALL LIST THE MANUFACTURER'S TEST NUMBER. TWO COPIES OF THE QUALIFICATION TEST REPORT, TOGETHER WITH THE DATA AND MATERIAL SPECIFIED ABOVE AND IN MIL-T-5041, SHALL BE FORWARDED TO THE NAVAL AIR SYSTEMS COMMAND WASHINGTON, DC 20361 ATTENTION: AIR-530321

NOTES:

1. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATION FOR BIDS, OR REQUEST FOR PROPOSAL EXCEPT THAT REFERENCE ADOPTED INDUSTRY STANDARDS SHALL GIVE THE DATE OF THE ISSUE ADOPTED.
2. FOR DESIGN FEATURE PURPOSES THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.

APPROVED 15 SEP 1964 REVISED (B) FOR CHANGES SEE SHEETS 1

This military standard is approved by NAVAL AIR SYSTEMS COMMAND, Department of the Navy and should be used by Navy activity. All other military activities are required to employ this standard where suitable.

P.A. NAVY - AS Other Cust	TITLE TIRE, PNEUMATIC, AIRCRAFT 36x11 HIGH SPEED, TYPE VII (NAVY)	MILITARY STANDARD MS90346(AS)
PROCUREMENT SPECIFICATION	SUPERSEDES:	SHEET 2 OF 2

DD FORM 1 MAR 72 672-1 (Limited coordination) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE. PLATE NO. 23071

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